# Cyber Security

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

## Fundamental IT Skills

- Computer Hardware Components
- Connection Types and their function
  - NFC
  - Bluetooth
  - WiFi
  - Infrared
- OS-Independent Troubleshooting
- Understand Basics of Popular Suites
  - MS Office Suite
  - iCloud
  - Google Suite
- Basics of Computer Networking

### CTFs (Capture the Flag)

- HackTheBox
- TryHackMe
- VulnHub
- picoCTF
- SANS Holiday Hack Challenge

### Certifications

Beginner Certifications

- CompTIA A+
- CompTIA Linux+
- CompTIA Network+
- CCNA
- CompTIA Security+

Advanced Certifications

- CEH
- CISA
- CISM
- GSEC
- GPEN
- GWAPT
- GIAC
- OSCP
- CREST
- CISSP

## Operating Systems

- Windows
- Linux
- MacOS

Learn following for each

- Installation and Configuration
- Different Versions and Differences
- Navigating using GUI and CLI
- Understand Permissions
- Installing Software and Applications
- Performing CRUD on Files
- Troubleshooting
- Common Commands

## Networking Knowledge

- Understand the OSI Model
- Common Protocols and their Uses
- Common Ports and their Uses
- SSL and TLS Basics
- Basics of NAS and SAN

- Basics of Subnetting
- Public vs Private IP Addresses

IP Terminology

- localhost
- loopback
- CIDR
- subnet mask
- default gateway

Understand the Terminology

- VLAN
- DMZ
- ARP
- VM
- DHCP
- DNS
- NAT
- IP
- Router
- Switch
- VPN

- MAN
- LAN
- WAN
- WLAN

Understand these

- DHCP
- DNS
- NTP
- IPAM

Functions of each

- Star
- Ring
- Mesh
- Bus

Network Topologies

- SSH
- RDP
- FTP
- SFTP
- HTTP / HTTPS
- SSL / TLS

Network Protocols

- VMWare
- VirtualBox
- esxi
- proxmox

Common Virtualization Technologies

- Hypervisor
- GuestOS
- HostOS
- VM

Basics of Virtualization

- ipconfig
- ping
- dig
- netstat
- route
- nmap
- tcpdump
- arp
- tracert
- nslookup
- iptables
- Packet Sniffers
- Port Scanners
- Protocol Analyzers

Troubleshooting Tools

- Kerberos
- RADIUS
- LDAP
- SSO
- Certificates
- Local Auth

Authentication Methodologies

## Security Skills and Knowledge

- Understand Common Hacking Tools
- Understand Common Exploit Frameworks
- Understand Concept of Defense in Depth
- Understand Concept of Runbooks
- Understand Basics of Forensics
- Basics and Concepts of Threat Hunting
- Basics of Vulnerability Management
- Basics of Reverse Engineering
- Penetration Testing Rules of Engagement
- Perimiter vs DMZ vs Segmentation

- Core Concepts of Zero Trust
- Roles of Compliance and Auditors
- Understand the Definition of Risk
- Understand Backups and Resiliency
- Cyber Kill Chain
- MFA & 2FA
- Honeypots
- Operating System Hardening
- Understand Concept of Isolation
- Basics of IDS and IPS
- Authentication vs Authorization

- Blue / Red / Purple Teams
- False Negative / False Positive
- True Negative / True Positive
- Basics of Threat Intel, OSINT
- Understand Handshakes
- Understand CIA Triad
- Privilege Escalation
- Web Based Attacks and OWASP10
- Learn how Malware works and Types

### Tools for Incident Response and Discovery

- dig
- nmap
- ping
- arp
- cat
- dd
- tail
- hping
- head
- grep
- nslookup
- tracert
- winhex
- autopsy
- ipconfig
- curl
- wireshark
- memdump
- FTK Imager

### Understand Common Standards

- ISO
- RMF
- NIST
- CIS
- CSF

### Using tools for Unintended Purposes

- LOLBAS
- GTFOBINS
- WADCOMS

### Learn how to find and use these logs

- Event Logs
- syslogs
- netflow
- Packet Captures
- Firewall Logs

### Understand Hardening Concepts

- MAC-based
- NAC-based
- Port Blocking
- Group Policy
- Sinkholes
- ACLs
- Patching
- Jump Server
- Endpoint Security

### Understand Common Tools

- VirusTotal
- urlscan
- any.run
- Joe Sandbox
- urlvoid
- WHOIS

### Understand Audience

- Stakeholders
- HR
- Legal
- Compliance
- Management

### Basics of Cryptography

- Salting
- Hashing
- Key Exchange
- Private vs Public Keys
- PKI
- Obfuscation

### Understand Frameworks

- Diamond Model
- Kill Chain
- ATT&CK

### Common Distros for hacking

- ParrotOS
- Kali Linux

### Understand the following

- SIEM
- SOAR

### Secure vs Unsecure Protocols

- FTP vs SFTP
- SSL vs TLS
- IPSEC
- DNSSEC
- LDAPS
- SRTP
- S/MIME

### Understand the following Terms

- Antivirus
- Antimalware
- EDR
- DLP
- Firewall & Nextgen Firewall
- HIPS
- NIDS
- NIPS
- Host Based Firewall
- Sandboxing
- EAP vs PEAP
- WPS
- ACL
- WPA vs WPA2 vs WPA3 vs WEP

### Understand the Incident Response Process

- Preparation
- Identification
- Containment
- Eradication
- Recovery
- Lessons Learned

### Understand Threat Classification

- Zero Day
- Known vs Unknown
- APT

### Attack Types and Differences

- Phishing
- Whishing
- Whaling
- Smishing
- Spam vs Spim
- Shoulder Surfing
- Tailgating
- Dumpster Diving
- Zero day
- Social Engineering
- Reconnaissance
- Impersonation
- Watering Hole Attack
- Drive by Attack
- Typo Squatting
- Brute Force vs Password Spray

### Common Attacks

- DoS vs DDoS
- MITM
- CSRF
- Spoofing
- SQL Injection
- XSS
- Evil Twin
- VLAN Hopping
- DNS Poisoning
- Deauth Attack
- Replay Attack
- Rogue Access Point
- Buffer Overflow
- Memory Leak
- Pass the Hash
- Directory Traversal

## Cloud Skills and Knowledge

- Understand the Concept of Security in the Cloud
- Understand the differences between cloud and on-premises
- Understand the concept of Infrastructure as Code
- Understand the Concept of Serverless
- Understand the basics and general flow of deploying in the cloud

### Understand Cloud Services

- SaaS
- PaaS
- IaaS

### Cloud Models

- Private
- Public
- Hybrid

### Common Cloud Environments

- AWS
- GCP
- Azure

### Common Cloud Storage

- S3
- Dropbox
- iCloud
- Box
- OneDrive
- Google Drive

## Programming Skills

- Python
- Go
- JavaScript
- C++
- Bash
- Power Shell

## Keep Learning